

Soft Actor-Critic (SAC)

1 Maximum Entropy Objective

Standard RL algorithms maximize the expected discounted return:

$$J(\pi) = \sum_{t=0}^{\infty} \mathbb{E}_{(s_t, a_t) \sim \rho_{\pi}} [r(s_t, a_t)] \quad (1)$$

In contrast, Soft Actor-Critic incorporates an **entropy term** to encourage exploration and robustness:

$$J(\pi) = \sum_{t=0}^{\infty} \mathbb{E}_{(s_t, a_t) \sim \rho_{\pi}} [r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot \mid s_t)))] \quad (2)$$

where α is the temperature parameter controlling the relative importance of entropy, and the policy entropy $\mathcal{H}(\cdot)$ is:

$$\mathcal{H}(\pi(\cdot \mid s_t)) = \mathbb{E}_{a_t \sim \pi(\cdot \mid s_t)} [-\log \pi(a_t \mid s_t)] \quad (3)$$

2 Soft Q-Function and Value Function

The **Soft Q-function** represents the expected return plus future expected entropy:

$$Q(s_t, a_t) = r(s_t, a_t) + \mathbb{E}_{(s_{t+k}, a_{t+k}) \sim \rho_{\pi}} \left[\sum_{k=1}^{T-t} \gamma^k (r(s_{t+k}, a_{t+k}) + \alpha \mathcal{H}(\pi(\cdot \mid s_{t+k}))) \right] \quad (4)$$

The relationship between $V(s_t)$ and $Q(s_t, a_t)$ in the soft framework is given by:

$$V(s_t) = \mathbb{E}_{a_t \sim \pi(\cdot \mid s_t)} [Q(s_t, a_t) - \alpha \log \pi(a_t \mid s_t)] \quad (5)$$

$$= \mathbb{E}_{a_t \sim \pi} [Q(s_t, a_t)] + \alpha \mathcal{H}(\pi(\cdot \mid s_t)) \quad (6)$$

3 Soft Policy Iteration Operator

Define the Soft Bellman backup operator $\mathcal{T}^{\pi}$ as:

$$\mathcal{T}^{\pi} Q(s_t, a_t) \triangleq r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1} \sim p, a_{t+1} \sim \pi} [Q(s_{t+1}, a_{t+1}) - \alpha \log \pi(a_{t+1} \mid s_{t+1})] \quad (7)$$

Convergence Lemma: For any bounded soft Q-function $Q^0 : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, defining $Q^{k+1} = \mathcal{T}^{\pi} Q^k$, the sequence Q^k converges to the soft Q-value of policy π as $k \rightarrow \infty$.

4 Off-Policy Learning and Critic Update

In model-free off-policy learning, transitions (s_t, a_t, r_t, s_{t+1}) are sampled from a replay buffer $\mathcal{D}$. The target value is approximated via a single-sample Monte Carlo estimator:

$$y_{\text{ideal}} = r_t + \gamma (Q(s_{t+1}, a_{t+1}) - \alpha \log \pi_{\phi}(a_{t+1} \mid s_{t+1})), \quad a_{t+1} \sim \pi_{\phi}(\cdot \mid s_{t+1}) \quad (8)$$

To mitigate overestimation bias, SAC employs **Clipped Double-Q Networks** ($Q_{\theta_1}, Q_{\theta_2}$) and target networks ($\bar{\theta}_1, \bar{\theta}_2$) updated via exponential moving average $\bar{\theta} \leftarrow \tau \theta + (1 - \tau) \bar{\theta}$.

The practical target y_t is computed as:

$$y_t = r(s_t, a_t) + \gamma \left(\min_{i=1,2} Q_{\bar{\theta}_i}(s_{t+1}, a_{t+1}) - \alpha \log \pi_{\phi}(a_{t+1} \mid s_{t+1}) \right), \quad a_{t+1} \sim \pi_{\phi}(\cdot \mid s_{t+1}) \quad (9)$$

Each Critic network Q_{θ_i} is updated by minimizing the Mean Squared Error (MSE) loss:

$$J_Q(\theta_i) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}) \sim \mathcal{D}} \left[(Q_{\theta_i}(s_t, a_t) - y_t)^2 \right] \quad (10)$$